GERMÁN KÜBER

Staff AI Engineer — LLM Systems, Multi-Agent Architectures, Real-Time AI

Las Palmas, Spain · EU Citizen — authorized to work across the EU/EEA, no visa sponsorship required
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PROFESSIONAL SUMMARY

Staff-level AI engineer with 15+ years building production distributed systems and recent focus on shipping production LLM systems, agentic workflows, multi-agent systems, and real-time AI. Shipped and operated multi-provider LLM orchestration spanning **OpenAI, Anthropic, Google Gemini**, plus managed cloud services (AWS Bedrock, Azure OpenAI Service, Google Vertex AI), with dynamic model routing for cost and performance optimization. Hands-on experience implementing MCP (Model Context Protocol) servers and clients, function calling and structured outputs (JSON schema validation), and ReAct-style agent patterns. Built production RAG pipelines across Pinecone, Chroma, pgvector, and Redis with embeddings, semantic chunking, reranking and hybrid retrieval, and instrumented a full LLM evaluation and observability stack (LangSmith, RAGAS, DeepEval, Langfuse, OpenTelemetry GenAI conventions). Designed and ran systematic eval safety and red-teaming frameworks, significantly reducing hallucination rate through structured evals. Real-time voice AI experience includes pipelines integrating OpenAI Realtime API, Deepgram, and ElevenLabs with sub-200ms TTFT (Time To First Token). Cut combined LLM and cloud spend by 45% through semantic caching, prompt compression, and dynamic model selection. Polyglot across **Python, TypeScript / Node.js, .NET and Rust**. EU citizen (Spanish nationality) with full authorization to work across the EU/EEA — no visa or sponsorship required — open to self-funded relocation anywhere in the EU.

CORE EXPERTISE

AI / LLM Systems: Multi-provider LLM orchestration (OpenAI, Anthropic Claude, Google Gemini, Llama), AWS Bedrock, Azure OpenAI Service, Google Vertex AI, dynamic model routing, prompt engineering, context engineering, structured outputs (JSON schema validation), function calling / tool calling, cost optimization (semantic caching, prompt compression, model selection), fine-tuning experimentation (LoRA / QLoRA)

Multi-Agent & Agent Engineering: agentic workflows, LangChain v1, LangGraph, multi-agent orchestration (supervisor pattern, conditional routing), ReAct pattern, state machines, tool calling, function calling, human-in-the-loop, durable execution, MCP (Model Context Protocol) — server and client implementation

Vector Databases & RAG: Pinecone, Chroma, pgvector, Redis (RediSearch), PostgreSQL, semantic search, hybrid retrieval, chunking strategies, semantic chunking, embeddings, embedding models, reranking (cross-encoders), retrieval recall@k, production-grade RAG pipelines

Real-Time AI & Voice: OpenAI Realtime API, Deepgram (STT), ElevenLabs (TTS), SignalR, WebSocket streaming, real-time audio pipelines, sub-200ms TTFT (Time To First Token), voice activity detection, barge-in handling

AI Safety, Guardrails & Red-Teaming: Systematic eval safety frameworks, red-teaming pipelines, hallucination mitigation, output guardrails, content filtering, JSON schema validation for structured outputs, adversarial testing

LLM Evaluation & Observability: LangSmith, RAGAS, DeepEval, Langfuse, OpenTelemetry GenAI semantic conventions, evaluation frameworks (faithfulness, correctness, answer relevancy), agent task completion rate, hallucination rate measurement, trace analysis, regression testing, production monitoring

Programming Languages: Python (FastAPI, async), TypeScript / Node.js, JavaScript, C# / .NET, Rust, Cython, SQL

Cloud & AI Infrastructure: AWS (Bedrock, SageMaker, Lambda, S3, ECS), Azure (OpenAI Service, AI Search), Google Cloud (Vertex AI), Docker, Kubernetes, FastAPI, CI / CD, Prometheus, observability stacks

ML / Inference: PyTorch, Ollama (local inference), Hugging Face

Distributed Systems: Async runtimes, low-latency networking, fault tolerance, microservices, event-driven architectures, high-throughput pipelines, real-time signal processing

Programming Paradigms & Design: Object-Oriented Programming (OOP), Functional Programming, design patterns, architecture patterns, data structures, algorithm design, SOLID principles, clean code

Engineering Practices & Leadership: Test-Driven Development (TDD), pair programming, code review, mentoring, technical interviewing, ADRs, design reviews, CI / CD, DevOps practices, Infrastructure as Code (Terraform), technical roadmap, on-call and incident response

PROFESSIONAL EXPERIENCE

Lead AI Architect & Co-Founder

KliverAI (Own Venture, closed Feb 2026) — Mar 2025 – Feb 2026 (full-time) — Remote, Spain

- Shipped multi-provider LLM orchestration integrating OpenAI, Anthropic Claude, and Google Gemini through both direct APIs and managed cloud services (AWS Bedrock, Azure OpenAI Service, Google Vertex AI), with dynamic model routing to pick the best model per task and cut inference spend.
- Rewrote the real-time layer from app-embedded SignalR to Azure SignalR Service with Redis-backed state, scaling concurrent users by 500% while holding sub-200ms TTFT on voice pipelines (OpenAI Realtime API).
- Benchmarked TTS / STT services (ElevenLabs, Deepgram, AssemblyAI, OpenAI Whisper, Play.ht) on latency, cost, and quality — getting latency to 90ms and word accuracy to nearly 95%.
- Built an adaptive simulation engine with multi-agent architecture on LangGraph, including state machines, conditional routing, ReAct-style reasoning, multi-step branching logic, and RAG-powered personalization for training scenarios.
- Implemented function calling and structured outputs (JSON schema validation) for reliable agent-tool integration across production workflows.
- Shipped production RAG pipelines across Pinecone, Chroma, pgvector, and Redis (RediSearch), with semantic search, hybrid retrieval, reranking via cross-encoders, and chunking strategies tuned per use case.
- Architected real-time voice AI pipelines integrating OpenAI Realtime API, Deepgram (STT), ElevenLabs (TTS), and SignalR, achieving sub-200ms TTFT across concurrent user sessions.
- Implemented MCP (Model Context Protocol) servers and clients to standardize tool access and context integration across LLM-powered workflows.
- Designed a Zero-Knowledge proof system using Noir, Barretenberg, and Garaga on StarkNet for verifiable expert credentials in a decentralized evaluation marketplace.
- Designed and ran systematic eval safety and red-teaming frameworks, significantly reducing hallucination rate through structured evals across critical paths.
- Instrumented LLM evaluation and observability stack using LangSmith, RAGAS, DeepEval, Langfuse, and OpenTelemetry GenAI semantic conventions for regression testing, faithfulness / correctness scoring, and production monitoring.
- Owned technical strategy, architectural decisions, and delivery as the senior-most engineer; led platform development end-to-end on a polyglot stack (Node.js / TypeScript, Python, .NET, Rust, Kubernetes, AWS). Hired and led a distributed team of up to 3 engineers.

Fractional CTO (Part-time, hands-on)

Rocking Product — Apr 2025 – Dec 2025 (part-time, parallel to KliverAI) — Remote

- Came in as Fractional CTO to an 8-engineer team and restructured it in the first quarter without stepping away from the code; ran technical interviews and hiring decisions.
- Built an image embeddings service (Azure AI Vision and CLIP) to improve product matching across brand comparisons, raising accuracy and lifting the client's revenue by 40%.
- Cut production bugs by 70% through team training and automated processes (testing, CI / CD, code review).
- Benchmarked LLM usage and optimized cloud infrastructure through semantic caching, prompt compression, and dynamic model selection across providers, cutting combined API and cloud spend by 45%.
- Designed a reusable base architecture on LangChain and LangGraph with monitoring via Langfuse and LangSmith and prompt versioning; migrated the real-time LLM service from OpenAI to Azure AI Foundry for compliance and data retention, cutting latency and improving scalability.
- Introduced Spec-Driven Development (SDD) and AI-driven development workflows with autonomous agents automating code review, testing, and deployment.
- Shipped AI-powered features in Node.js / TypeScript and Python (FastAPI, LangChain), instrumented the team's first LLM evaluation and monitoring with LangSmith and Langfuse, and ran weekly public-speaking workshops to sharpen team communication.

Lead Software Engineer (Distributed Systems / Blockchain)

Borderless Capital / CTF Capital — Sep 2021 – Mar 2025 — Remote

- Lead engineer for low-latency distributed systems in Rust (tokio, async) and Python / Cython; pipelines processed 10K+ opportunities per day from $50M+ in daily on-chain flow and contributed $100K+ in attributable annual P&L with execution under 10ms.
- Migrated the system from Python / Cython to Rust and took it to 20x more transactions per second.
- Built high-throughput data pipelines and real-time signal processing across multi-chain ecosystems (EVM, Solana), with custom transaction delivery infrastructure achieving sub-slot inclusion on Solana (~400ms); reverse-engineered Geth and Reth to improve transaction simulation and gas estimation, contributing to the REVM community.
- Owned 24/7 production reliability and incident response, including on-call rotation, post-mortems, and observability ownership with Prometheus and custom Rust telemetry tracking 50+ real-time metrics.
- Ran technical due diligence on 5+ projects that ended up getting funded — architecture, risk, and feasibility — acting as the CTO's technical reference on the low-level side.
- Acted as technical lead across the blockchain team, which grew from 4 to 10 engineers: code review, mentoring, technical interviewing, and architectural decisions.

Senior Software Architect
Paramo Technology — Oct 2020 – Sep 2021 — Remote

- Owned the end-to-end architecture of a high-traffic online casino (iGaming) platform handling real-time money operations and regulatory reporting, working directly with the CTO.
- Spotted the issues slowing teams down and designed action plans that raised productivity by 40%, measured by features shipped and fewer errors.
- Designed and rolled out event-driven microservices in .NET (C#, ASP.NET Core) for real-time game state, wallet operations, and regulatory reporting; migrated relational databases to MongoDB where it made sense.
- Implemented distributed tracing across all services (OpenTelemetry, Application Insights, Datadog) to catch bugs early and resolve production issues faster.
- Brought in Kubernetes to scale; built automated deploy pipelines; helped migrate services to newer .NET versions.
- Led part of the hiring and designed the technical entry exam; created and delivered internal training programs on architecture patterns, design patterns, testing discipline, and operational best practices.

Software Architect
CloudX (Software Consultancy) — Sep 2017 – Oct 2020 — Remote / Buenos Aires (first employee)

- Was the company's first employee; led architecture decisions across several enterprise product lines, often working on-site with clients, and helped build part of the team as it grew.
- Designed Angular and React frontends with reusable component libraries and Node.js / TypeScript and .NET microservices backends using Domain-Driven Design.
- Built a distributed system over geospatial data with PostGIS for a UK client.
- Delivered technical training and conference talks to partners and internal teams.

Software Architect (Angular / .NET)
Majestic Hotel Group — Jan 2014 – Aug 2017 — Argentina

- Designed and shipped full-stack features in Angular and .NET for internal hotel management platforms, first as Team Lead and later as architect; led the migration from AngularJS to Angular 2.
- Redesigned the data model toward an event-oriented architecture, reducing coupling across integrations.

.NET Web Developer
Plus54 — 2011 – 2014 — Buenos Aires (earlier: Acsys 2009–2011, IBM 2008–2009)

- Early career: built .NET web applications, internal tooling, and mobile components end-to-end; integrated external APIs with unit testing and reliable data exchange patterns.

KEY ACHIEVEMENTS

Real-Time AI Platform at KliverAI — Multi-provider LLM orchestration across OpenAI, Anthropic, Gemini, AWS Bedrock, Azure OpenAI, and Vertex AI, with sub-200ms voice latency and full evaluation/observability stack.
45% LLM Cost Reduction at Rocking Product — Cut combined API and cloud spend through semantic caching, prompt compression, and dynamic model routing in production.
Microsoft MVP, Developer Technologies — 8 consecutive years (2017–present), recognized for technical content and community contributions.
80+ International Technical Talks — .NET Conf, DevFest, NetCoreConf, vOpen, Global Azure Bootcamp, and meetups across LATAM and Europe.
Net-Baires Founder — Founded and grew Argentina's largest .NET developer community to 100K+ members across 10+ years.
Polkadot Builder Program Graduate — Completed Polkadot Blockchain Academy in Lisbon, Portugal (2026).
Microsoft Student Partner — 5 years (2016–2021).

EDUCATION & CERTIFICATIONS

Microsoft Certified: Azure Solutions Architect Expert (AZ-305) — Microsoft
Microsoft Certified: Azure Associate (AZ-104 / AZ-204 level) — Microsoft
Microsoft MVP, Developer Technologies — Feb 2017 – Present (8 consecutive years)
Polkadot Builder Program — Polkadot Blockchain Academy, Lisbon, Portugal (2026)
Systems Engineer — Universidad Abierta Interamericana (UAI), Buenos Aires, Argentina (2017–2021)

OPEN SOURCE & COMMUNITY LEADERSHIP

Net-Baires (.NET Community) — Founder (2015–present). Argentina's largest .NET developer community, 100K+ members, 10+ years of monthly meetups and conferences.
Crypto Para Mortales — Founder (2023–present). Spanish-language blockchain and crypto education conference.
Patrones Para Mortales — Founder (2020–present). Educational content brand on software design patterns and architecture.
YouTube channel (youtube.com/@germankuber) — technical deep-dives on Rust, blockchain, AI, and software architecture.

SPOKEN LANGUAGES

Spanish: Native — English: Full Professional Proficiency — daily use in international remote teams